1. Maximum Depth of Binary Tree (Easy)

Problem:
Given the root of a binary tree, return its maximum depth.

The maximum depth is the number of nodes along the longest path from the root to a leaf node.

Example 1

Input

root = [3,9,20,null,null,15,7]

Output

3

Example 2

Input

root = [1,null,2]

Output

2

Constraints

- Number of nodes: 0 to 10^4
- $-100 \le$ Node.val ≤ 100

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2. Same Tree (Easy)

Problem:
Given the roots of two binary trees p and q, check whether they are the same.

Two trees are the same if they have the same structure and the same node values.

Example 1

Input

p = [1,2,3]
q = [1,2,3]

Output

true

Example 2

Input

p = [1,2]
q = [1,null,2]

Output

false

Constraints

- Number of nodes: 0 to 100

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3. Invert Binary Tree (Easy)

Problem:
Given the root of a binary tree, invert the tree and return its root.

Example

Input

root = [4,2,7,1,3,6,9]

Output

[4,7,2,9,6,3,1]

Constraints

- Number of nodes: 0 to 100

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4. Symmetric Tree (Easy)

Problem:
Given the root of a binary tree, check whether it is symmetric around its center.

Example 1

Input

root = [1,2,2,3,4,4,3]

Output

true

Example 2

Input

root = [1,2,2,null,3,null,3]

Output

false

Constraints

- Number of nodes: 1 to 1000